



February 20th, 2026

HHS Health Sector AI RFI

➔ Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care

Submitted to:

**Department of Health and
Human Services**

Assistant Secretary for Technology Policy/
Office of the National Coordinator for
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February 20th, 2026

Steven Posnack
Principal Deputy Assistant Secretary for Technology Policy
330 C Street SW
Washington, DC 20201

Subject: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care

Dear Mr. Posnack:

I am pleased to submit ICF's response to the Department of Health and Human Services (HHS) Request for Information (RFI): Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care.

ICF is a global consulting and technology firm with more than five decades of experience supporting federal health agencies in modernizing programs, data, and digital systems. Our interdisciplinary teams span policy, technology, analytics, and human-centered design and deliver integrated, practical solutions that strengthen performance and outcomes across CMS, NIH, and other HHS divisions.

We bring direct experience translating policy into operational requirements, governance structures, measurable outcomes, and scalable technical approaches. Our work with CMS provides deep familiarity with program and payment levers that can incentivize safe clinical AI adoption. We develop certifiable and auditable frameworks that providers and vendors can implement to ensure trustworthy AI use.

ICF supports agencies in advancing robust AI evaluation methods through layered testing aligned with NIST principles, real-world monitoring, and data infrastructure that enables validation over time. We also help expand interoperability and standards adoption to unlock high-quality datasets and enable scalable deployment, always grounded in human-centered design and clinician-focused usability considerations.

We welcome the opportunity to help HHS address policy, reimbursement, and research considerations and accelerate safe, effective, and equitable clinical AI adoption.

Sincerely,

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Executive Summary

ICF (NASDAQ: ICFI) is a global consulting and technology services company with more than 50 years of successful performance helping public and private sector organizations navigate change and shape the future. Founded in 1969, ICF combines deep domain expertise in areas such as health, social programs, energy, environment, and digital modernization with advanced analytics, data science, and creative and digital engagement capabilities. Today, ICF is a publicly traded corporation with approximately 9,000 full-time and part-time employees worldwide and annual revenues of about \$2.0 billion.



ICF's interdisciplinary teams—policy specialists, technologists, data scientists, and creatives—work together to deliver integrated solutions that are practical, evidence-based, and measurable. Clients rely on ICF for high-stakes, mission-critical work that requires both subject-matter depth and rigorous program management, supported by mature quality, information security, and IT service management systems.

ICF brings a long-standing federal footprint, with expertise in federal health, supporting agencies as they modernize data, technology, and programs that improve outcomes, affordability, and equity. Across HHS and its operating divisions, ICF delivers end-to-end capabilities spanning strategy and policy, research and evaluation, digital modernization, data platforms and analytics, and implementation support for large-scale programs and systems. Our federal health work includes sustained support to CMS on enterprise solutions and modernization efforts and biomedical/public health data and modernization support for NIH/NLM, complemented by broader federal health offerings that strengthen surveillance, community programs, and health communications.

This RFI response focuses on the questions where ICF has specific, demonstrable experience and expertise: translating policy intent into operational requirements, implementable governance, measurable outcomes, and scalable technology approaches. In particular, our recommendations reflect practical experience with (i) payment and program levers that can be operationalized through CMS quality programs and models, (ii) certifiable requirements and auditable controls that can be implemented by providers and vendors, and (iii) evaluation and human-centered methods that increase clinical usability, trust, and adoption over time.

The remainder of this document recommends that HHS accelerate safe and effective clinical AI adoption by using proven CMS mechanisms to create meaningful incentives (e.g., QPP/PI measures, Improvement Activities, and MVP alignment) and by embedding outcome-based “pay for validated value” approaches within high-leverage CMMI models; establish a voluntary certification and contracting/procurement baseline for non-medical AI used in clinical settings; advance evaluation methods that combine NIST-aligned risk management with layered testing (model outputs, system artifacts, and real-world user experience) and robust post-deployment monitoring supported by gold-standard datasets, SBIR/prizes, and standards; and expand interoperability through consistent adoption of key health data standards to enable benchmark datasets, portability, and scalable deployment—while prioritizing HCD, trustworthy AI, and



human–AI interaction research to ensure solutions fit clinical workflows and improve patient experience.

Incentivizing Effective Use of AI in Clinical Care (Q. #2)

HHS can and should incentivize the effective use of AI in clinical care. For example, HHS can use the **Quality Payment Program (QPP)** and replicate the **CEHRT/Promoting Interoperability (PI) playbook by setting a clear, testable “floor” for what qualifies as acceptable technology and use and then attaching meaningful scoring and payment consequences.**

Specifically, CMS should do the following:

1. Add a QPP definition for a certifiable class of clinical AI (e.g., “certified clinical AI module”) by building on the existing CEHRT definitional structure in 42 CFR § 414.1305 and aligning certification mechanics to the ONC Health IT Certification Program authorities in 45 CFR part 170, subpart E.
2. Update the PI performance category at 42 CFR § 414.1375 to create a scored objective/measure set for “safe and effective clinical AI use,” limited to certified modules and requiring auditable basics (e.g., version transparency/audit logs, clinician override, performance monitoring). This will ensure CMS pays for verified capability rather than marketing claims.
3. Add AI governance and post-deployment monitoring as high-weight Improvement Activities via the IA inventory process at 42 CFR § 414.1355.
4. Prevent fragmented incentives by establishing one or more “AI-enabled care” MIPS Value Pathways (MVPs) under 42 CFR § 414.1365 that bundle the AI expectations across Quality/Cost/IA/PI instead of scattering optional attestations.

In addition, because the CEHRT approach also flowed into Advanced APM design, CMS should extend the same logic to Advanced APM alignment by building on the existing Advanced APM CEHRT criterion at 42 CFR § 414.1415—not by requiring “AI” generically, but by requiring that any high-impact AI used in clinical decision support or automation within an Advanced APM context meet certifiable, auditable, monitored standards. This is a direct analogue to how “use CEHRT” is operationalized today. Together, these steps will allow HHS to incentivize effective AI use in clinical care using an established program and proven mechanisms.

HHS can also use CMMI Innovation Models to incentivize AI use in clinical care settings. To do so, HHS should **prioritize programmatic design changes that (a) lower the upfront adoption barrier the way CEHRT programs did, but (b) only “pay for AI” when it produces validated value and remains safe in real-world use.** The fastest way to do this would be to hardwire an “AI capability track” into high-leverage models. LEAD is explicitly positioned to launch after ACO REACH concludes at the end of 2026, making it the cleanest successor vehicle for model-wide expectations (e.g., allowing infrastructure-style payments or enhanced care management payments for certified AI deployment while gating shared savings/benchmarks on monitored performance and equity checks).

Similarly, models with longitudinal care management or defined episodes can incorporate AI in tightly measured ways. For example, Kidney Care Choices (KCC) has been extended through 2027, so CMS can add a standardized option to support certified AI-enabled risk



stratification/care management as a covered “model capability,” while requiring ongoing monitoring and evaluation tied to model endpoints. The Enhancing Oncology Model (EOM) began July 1, 2023 and added a second cohort July 1, 2025, creating a ready platform for AI-enabled symptom monitoring and care navigation. As noted above, this will work best when paired with post-deployment monitoring requirements and outcome-based gates rather than pure technology attestation. Because TEAM is explicitly a mandatory episode-based model, CMS can require transparency and auditability safeguards whenever participants use AI for episode identification, discharge planning, or post-acute decision support—protecting beneficiaries while still encouraging adoption through standardized requirements rather than ad hoc vendor-by-vendor negotiations.

By incentivizing AI use with these models, HHS can leverage existing processes and mechanisms to encourage AI adoption and use, accelerating innovation and maximizing benefit to citizens and stakeholders.

Novel Issues with AI in Non-Medical Devices (Q. #3)

It is true that use of AI in non-medical devices in clinical care settings may raise novel legal and implementation issues. AI technologies are rapidly evolving, and gaps are emerging in existing frameworks governing liability, privacy, safety, and accountability. However, HHS is well-positioned—through the OneHHS approach articulated in its recent AI Strategy—to lead by clarifying expectations, enabling innovation, and safeguarding patients. We recommend two priority actions.

First, HHS should **establish a voluntary certification framework for non-medical AI tools**, modeled on ONC’s Health IT Certification Program. This framework would accelerate the responsible adoption AI tools while reducing the regulatory burden on innovators by providing a means to evaluate and compare:

- Transparency and documentation.
- Safety monitoring and performance metrics.
- Bias and fairness assessments.
- Data protection and access safeguards.

Second, HHS should **issue model contracting and procurement guidance**. By building off OMB Memorandum M-25-22 and the HHS AI Strategy’s procurement guidance, especially the use of “Cross-Cutting Enablers,” HHS should provide health systems and vendors a consistent baseline by publishing contractual expectations for:

- Model auditability
- Data rights and ownership
- Security and privacy protections
- Prohibitions on unauthorized model training
- Clear user responsibilities

By taking these steps, HHS can provide the necessary guardrails to enable effective and safe innovation with AI in non-medical devices for clinical care settings.



Most Promising AI Evaluation Methods (Q. #4)

There are several promising approaches for evaluating AI in non-medical devices and we believe HHS should support them, especially because the best methods for ensuring safe, reliable operation with LLM-powered AI solutions are currently subject to research and debate. Further work is required to devise the best approaches for understanding, testing, and evaluating these technologies, especially in clinical care settings. HHS should take the lead by establishing standards and supporting research and investment. Doing so will increase trust, improve performance, and accelerate adoption and use, thereby bringing benefit to providers and patients.

NIST's AI Risk Management Framework (RMF) is a useful lens for orienting this work, as its four functions (Govern, Map, Measure, Manage) provide a means of assessing LLM performance and the risks of inadequate evaluation mechanisms. Another useful lens is the user. We believe focusing on users will be the best way to ensure evaluation mechanisms align with desired outcomes. Care providers are likely to be the greatest obstacle to broad adoption in clinical care settings. Accordingly, HHS should use Human Centered Design (HCD) to position care providers and patients at the center of evaluation method design and development. In addition, when promoting adoption, HHS should encourage an incremental, evolutionary approach that will create successes and build confidence. If done well, this will create a flywheel effect that builds momentum. Initial successes will build confidence, leading to further adoption and additional successes. Adoption and use will expand from there.

Promising Evaluation Methods

A variety of AI evaluation methods should be used to support increased adoption and use. Currently, some of the most promising approaches include the following:

- Employing LLMs as judges.
- Adjusting LLM training approaches penalize hallucinations.
- Using dynamic evaluation methods to adjust benchmarks as LLM capabilities improve, so that progress is consistent.
- Employing observability with tools like Langfuse.
- Red teaming to discover unanticipated LLM vulnerabilities.

Each of these approaches merits further research and support, especially in the field of clinical care. We recommend integrating HCD with all these approaches, so that the behavior of AI solutions is evaluated comprehensively and friction in the human-machine interface is proactively identified and addressed.

Our concept follows the effective pattern of layering AI evaluations to provide a comprehensive assessment of a solution's effectiveness. These layers typically include evaluating LLM responses in isolation, evaluating artifacts created from those responses by AI solutions (the human-machine teaming that embeds the LLM), and evaluating the experiences that those responses, artifacts, and solutions create. This layered approach is especially important in clinical care. Both patients and care providers will interact with and experience an AI solution. If those experiences are positive and valuable, then AI can drive improved outcomes for everyone.



Evaluation Metrics

There are several important metrics to consider for evaluation. Accuracy is crucial. Given the nature of LLMs, accuracy is best broken into components, such as consistency, factual correctness, and error count. Responses must also be relevant; LLMs should be evaluated on how well they answered the question and whether they accessed the right data. This necessitates subjectivity; user feedback on relevance, for example, is essential as people currently assess relevancy far better than machines. This is one of the primary reasons why we contend it is so important to employ HCD principles and approaches. Tone and clarity are also important factors that require human assessment. Was the response easily understood or did it lead to confusion? Was it at the right reading level? Finally, is the output fact-based and unbiased? As OMB Memo M-26-04 states, it is important to ensure that AI solutions present truthful, ideological neutral outputs. Accordingly, evaluation methods must account for these factors.

Supporting the Development of Improved Evaluation Methods

To support the development of improved evaluation mechanisms, several mechanisms are available. First, HHS ought to develop and provide “gold standard” datasets for training and evaluation of AI systems in clinical care settings. Those datasets could be created by HHS itself, or HHS could sponsor their creation through grants, prize competitions, or contract work. Once standardized data for training and evaluation is in place, progress with evaluation methods will be more rapid and advance with greater confidence.

In addition, HHS should use SBIRs to encourage development of new evaluation methods and tools. Focused grants and prize competitions should explore how best to use LLMs-as-judges in specific clinical care domains. Grants and prizes should also be used to encourage the development of additional models and datasets well-tuned to medical needs. Grants should be issued for encouraging the development of automated mechanisms to dynamically keep those models and datasets up to date as the field advances, a current research focus that HHS ought to encourage. Grants should also be considered for developing effective red-teaming techniques for clinical care scenarios and for identifying the most suitable metrics for evaluating AI solutions in clinical care contexts.

HHS can also support and accelerate adoption by providing guidance and standards. How should humans be centered in the evaluation process? How can vendors and partners leverage HCD with confidence to integrate actual patients and clinicians into evaluations? HHS should encourage the development of standards for these kinds of human-AI interactions in clinical care settings, so that the industry can integrate human impacts into evaluation methods and practices. Standards should also include acceptable levels of trust and bias, the use of LLMs-as-judges, and how AI solutions should handle unanticipated failure modes when their use goes beyond the scope of existing evaluations. By taking these steps, HHS can have the greatest impact on AI solutions used for clinical care outside of medical devices.

HCD in Clinical Care AI

Human-Centered Design (HCD) builds reliability and trust in clinical AI by turning human interactions, recovery from errors, and real-life use situations into clear evaluation goals. Otherwise, they risk being left as untested assumptions. While model-level checks (like accuracy, hallucination rates and robustness) show if an AI system **can** do a task, HCD looks at whether the system works reliably **when used in clinical workflows**. For instance, usability



testing with clinicians can reveal if uncertainty is communicated clearly, if explanations make sense under time pressure, and if the interface helps clinicians use AI properly. Signals, like how often clinicians override the AI, make corrections, or hesitate, act as early signs of trust and reveal potential system failures that model-only evaluations might miss.

HCD complements other LLM evaluation methods by considering real-world context. While techniques like LLMs-as-judges or dynamic benchmarking can give scores to outputs, HCD methods check if those outputs are useful, easy to understand and fit for the specific clinical decision. For example, two models might get similar scores for factual accuracy, but HCD testing can reveal that one model offers explanations that are more likely to lead to misunderstanding of risk among clinicians. Usability testing with scenarios, participatory design sessions, and practice clinical walkthroughs let evaluators see how AI behaves in real situations (like during handoffs, interruptions, or with missing information) and make sure safety features—such as alerts, warnings, or escalation steps—work properly in real usage settings.

HCD also assists effective monitoring after deployment. Instead of just tracking technical results, HCD-based monitoring looks at how trust in the system changes over time. For example, HCD can catch whether clinicians start ignoring AI outputs, start trusting them too much, or changing their work in unexpected ways. These patterns can be tracked with simple feedback tools—like pop-up prompts in the software, regular surveys linked to specific AI uses, or clinician decisions to override or correct AI suggestions—and used with technical checks to know when to review the system. In this way, HCD acts as the final layer of a comprehensive evaluation strategy that helps ensure AI stays safe, helpful, and trusted as clinical settings, users, and the technology change.

To strengthen responsible clinical AI implementation, HHS should support HCD approaches that include intentional human-to-human engagement as a core component. Direct observation of clinicians who are using AI tools, paired with one-on-one interviews and small-group listening sessions, provides critical insight into how and why clinicians rely (or not) on these systems in real workflows. These interactions uncover tacit knowledge, context-specific decision cues, and early signs of misunderstanding or misuse that technical evaluations cannot detect. Just as importantly, involving clinicians in this way builds trust and accelerates adoption: they see their concerns reflected in design decisions, understand how their input influences implementation, and experience the tools as being shaped with them rather than delivered to them. By supporting HCD methods that prioritize these human touchpoints, HHS can help ensure clinical AI tools are implemented responsibly, safely, and in ways that genuinely support the clinicians who use them.

HHS Support for Private Sector Activities (Q. #5)

HHS can support private sector activities like accreditation, certification, industry-driven testing, and credentialing for clinical AI by adopting “tried-and-true” federal playbooks, such as:

1. Risk-based premarket review and lifecycle oversight (FDA).
2. Reusable third-party assurance that agencies can rely on (FedRAMP).
3. Scaled compliance pathways that mix self-attestation with independent certification for higher-risk products (FCC).
4. Rapid adoption enabled by clear flexibilities, aligned payment, and temporary enforcement discretion (COVID-era telehealth).



The common approach across these models is that government sets clear, testable expectations and then recognizes independent private-sector assurance artifacts, allowing providers and innovators to scale without every payer or program creating a unique and independent review. HHS can leverage all four.

First, HHS can mirror the FDA approach by defining risk tiers and evidence expectations for clinical AI and then formally recognizing qualified private sector entities that generate standardized “assurance packages”—such as testing results, validation protocols, usability evidence, and monitoring plans—that HHS programs can trust. FDA already operates multiple, well-understood pathways for medical devices grounded in the FD&C Act and implemented through established review practice. Examples include 510(k) clearance, De Novo classification for novel low–moderate risk devices, and Premarket Approval (PMA) for Class III devices. HHS could apply the same logic to clinical AI that sits outside traditional device boundaries by publishing standardized submission templates that cover items such as intended use, training/validation data provenance, performance limits, human oversight, post-deployment monitoring, etc. and then recognizing third-party testing/certification programs that produce evidence in those formats. This would create a predictable, scalable path for “trusted AI” eligibility in Medicare and Medicaid program use cases.

Second, HHS can emulate FedRAMP by establishing an “assess once, reuse many times” model for clinical AI assurance, where private assessors perform independent evaluations and HHS can reuse the results across programs. FedRAMP was established by OMB in 2011 and later codified in law through the FedRAMP provisions enacted in Public Law 117-263 (NDAA FY2023), Section 5921. FedRAMP capitalizes on scalable private sector assurance through Third-Party Assessment Organizations (3PAOs) that are independently assessed by the American Association for Laboratory Accreditation (A2LA). HHS can take the same approach for clinical AI by establishing a reusable authorization construct for AI used in clinical care that ensures standards regarding security, privacy, auditability, monitoring, and performance are met. HHS can then recognize accredited “clinical AI assessment organizations” and allow providers/payers to rely on those reusable authorizations to encourage adoption and use at scale.

Third, HHS can borrow from the FCC’s equipment authorization framework by creating two compliance tracks that support innovation while protecting patients: a lower-friction supplier attestation path for low-risk tools and a higher-rigor independent certification path for higher-impact clinical uses. FCC rules in 47 CFR part 2, subpart J, explicitly contemplate equipment being authorized under Supplier’s Declaration of Conformity or receiving a grant of certification (including via Telecommunication Certification Bodies), and these authorities are tied to statutory foundations including 47 U.S.C. § 302a. HHS could employ a similar concept, permitting streamlined attestation for limited-scope administrative AI (with audit rights and disclosures), while requiring independent certification (and potentially accredited labs) for AI that materially influences diagnosis, triage, treatment, or autonomous clinical actions. Adopting this approach would encourage consistency in labels, logs, and change-management disclosures so that certified tools could be regularly evaluated to ensure they remain trustworthy as they evolve.

Fourth, HHS can apply the COVID-era telehealth lesson: remove adoption blockers quickly, align payment, and clarify compliance expectations while using time-bound flexibilities and explicit evaluation to decide what becomes permanent. The Secretary’s emergency waiver authority under Social Security Act § 1135 (42 U.S.C. § 1320b–5) is designed to allow temporary waivers/modifications during emergencies. During COVID-19, CMS used this



posture to expand telehealth flexibility, noting that authority under the CARES Act broadened §1135 waiver authority and enabled additional telehealth waivers, including waivers tied to Medicare telehealth requirements and 42 CFR § 410.78 provisions. In parallel, HHS OCR issued the Notification of Enforcement Discretion for Telehealth Remote Communications under HIPAA during the nationwide public health emergency, reducing compliance friction for good-faith telehealth. HHS can adopt the same pattern for clinical AI by creating time-limited “AI adoption pilots” with clear guardrails to define what is allowed, aligned reimbursement and reporting to specify what is paid and measured, and targeted compliance flexibilities as and where appropriate to accelerate safe use. HHS could then use the collected evidence from these pilots to determine what aspects to institutionalize as durable accreditation/certification expectations.

Widening Opportunities through Interoperability (Q. #8)

Enhanced interoperability will widen market opportunities, advance clinical research, and accelerate the development and employment of AI for clinical care. AI use in healthcare will only be as effective as the consistency, quality, and portability of the data AI systems rely upon. Today, clinical data remains fragmented across EHR platforms, public health systems, research environments, claims systems, and social service data repositories, limiting the ability to create scalable, generalizable AI solutions that can operate reliably across populations and care settings. Additionally, data is often not “AI ready,” resulting unreliable, unsafe, biased, and non-portable outputs that undermine clinical trust and regulatory readiness.

Data Types

The greatest impact from interoperability will come from improving the clinical and contextual data used by AI systems and solutions. This includes EHR data (diagnosis, medications, lab results, etc.), imaging and diagnostics (DICOM) data, claims and utilization data, public health and surveillance data, genomic data, data from community resources, and data from personal devices. Incorporating unstructured clinical narratives and administrative metadata into available datasets will further strengthen AI’s ability to draw valuable inferences and support clinical care delivery. These data sources should be longitudinally linked so that AI can support predictive modeling, clinical decision support, assessments of population health, early interventions, and research at scale while enabling multimodal benchmarking and reproducible evaluations across representative cohorts.

Data Standards

Standards are essential for accelerating adoption and use of AI. We recommend consistent, broad adoption of HL7, FHIR, USCDI, DICOM, LOINC, and SNOMED standards to allow AI developers to build solutions that can operate across health systems and platforms without extensive data transformation. API-driven architectures will further allow AI tools to be embedded directly into clinical workflows rather than serving as standalone applications. These standards allow AI solution providers to build solutions once and deploy them across multiple EHRs, provider networks, research institutions, and public health agencies without costly data transformation for each environment. Effective use of standards will also reduce the need for custom integrations that would otherwise increase cost, limit competition, and slow innovation. By promoting standards in this way, HHS can lower barriers to entry for innovators and expand



the market for AI-enabled clinical tools that can be evaluated, procured, and implemented modularly to develop best-of-breed solutions.

Market Impact

Because AI effectiveness, safety, and reliability depend directly on the quality and structure of datasets used for solution development, validation, and deployment, interoperable data environments enable clinicians and researchers to confidently use AI tools in diagnostics, pediatric oncology, chronic disease research, and other high-stakes areas. When data is poorly organized, inconsistent, or incomplete, predictions become less accurate and clinical decision-making suffers. Interoperable systems directly address this by allowing normalization and semantic alignment across data sources, thereby maintaining strong data lineage and provenance and supporting transparent documentation. Interoperability also strengthens governance by enabling privacy-preserving de-identification, bias detection across diverse populations, and responsible data sharing. These elements streamline multi-omic and clinical data integration, support reproducible science, and ultimately expand opportunities for cross-institution research. Our recommended approach aligns with the HHS AI Strategy, especially the idea of cross-cutting enablers, as it will encourage clinical deployment of AI in ways that are aligned with federal strategies to reduce duplicative systems and strengthen enterprise-wide coordination.

Benchmarking

Enhanced interoperability will enable the creation and use of “gold standard” benchmark datasets that improve model validation across representative cohorts and care settings. We recommend the government either work to build such benchmark datasets or sponsor their development by establishing standards for development and evaluation of AI solutions in clinical care settings. HHS can ensure these datasets promote interoperability by requiring standardized structures, robust metadata, and privacy-preserving pipelines. This will ensure interoperability through by reducing model fragmentation and will prevent AI systems from being trained on narrow datasets that limit generalizability or introduce undesirable biases. By ensuring the use of consistent and validated datasets, HHS can increase the market viability for AI tools, accelerate evaluation and regulatory readiness, and strengthen confidence in real-world clinical use.

Specific Areas of AI Research to Prioritize (Q. #10)

AI’s true value will only be appreciated when it effectively augments human providers within clinical workflows. Today, most healthcare AI research focuses on algorithm accuracy in isolation, but ignores user experience, workflow integration, and human factors. This is unfortunate as these aspects are crucial for increased adoption and safe use. Accordingly, HHS should prioritize R&D that puts human users at the center of AI design and evaluation by embracing HCD. Two specific research areas stand out.

1. **Human-AI Interaction.** Focusing on human-AI interaction will address one of the greatest obstacles to adoption: clinician buy-in and workflow fit. If AI tools and solutions are intuitive, reduce human effort, and demonstrably enhance care delivery, providers are far more likely to embrace them. Conversely, even a highly accurate algorithm will fail in practice if it is difficult to use or disrupts proven clinical routines. By investing in this



research, HHS would accelerate the transition of AI from an impressive technology to a core component of trusted approaches that improve productivity and patient outcomes.

2. **Trustworthy AI.** Prioritizing Trustworthy AI research will help encourage adoption by reducing skepticism. Improved approaches to trustworthiness will provide the evidence, methods, and tools to create confidence that AI systems are safe, secure, and accountable. That will accelerate adoption by addressing doubt head-on. Clinicians will be more likely to trust AI recommendations if they know the system has been stress-tested for safety and bias, can explain itself, and is backed by governance that ensures issues are caught and addressed. Patients will be more receptive to AI-driven care if there's transparency and unbiased output. By investing in trustworthy AI R&D, HHS will foster the foundational “social license” for AI in healthcare—building public trust and providing the guardrails so innovation can proceed more rapidly without undue risk.

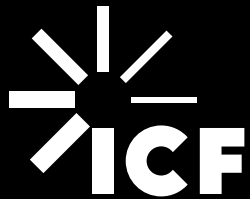
Ultimately, HHS should prioritize research investments that are designed to have a direct, positive impact on patients. Driving better patient outcomes is a clear goal. But we also recommend funding research designed to **improve patient experience**. Our nation's healthcare is the best in the world when we balance science with responsive, personalized care and clear, compassionate communication. Below is a broader summary of research areas worth further investment, aligned to patient experience.

Research Priorities for Clinical AI Adoption

Research Area and Relevance	Key Focus / References	Positive Impact on Patient Experience
Human-AI Interaction Ensure AI tools fit into clinical workflows and enhance (not hinder) healthcare professionals' performance and acceptance.	- Develop frameworks to measure & improve real-world human-AI collaboration in healthcare - Use HCD to align AI with user needs; prevent automation bias and user frustration	- More time with patients (less time entering data) - Clearer communication and explanations - Reduced frustration from workflow delays or errors - Greater confidence of being heard and understood
Trustworthy AI (Safety, Security, and Reliability) Builds a foundation of trust necessary for widespread adoption. If AI systems are demonstrably safe, secure, transparent, and accountable, stakeholders will be confident using them in care.	- Focus on reliability (robustness, adversarial defense), explainability, bias mitigation, and post-deployment monitoring - Align with NIST AI RMF and FUTURE-AI principles	- Confidence in safety and fairness of care - Transparent, understandable AI decisions - Reduced risk of harm or misdiagnosis - Equitable treatment across populations
Reinforcement Learning (RL) and Adaptive AI Enables AI to continuously improve and personalize in complex clinical decision scenarios (e.g., dynamic treatment planning), potentially boosting outcomes beyond static protocols.	- Use RL to optimize sequential decisions and personalize care - Develop safe RL techniques (offline training, human-in-loop oversight)	- More personalized, responsive care - Better long-term outcomes through adaptive treatment - Reduced trial-and-error in care plans



Research Area and Relevance	Key Focus / References	Positive Impact on Patient Experience
Collaborative & Multi-Agent Systems Ensures AI can operate in the complex multi-stakeholder environment of healthcare. Research here yields AI that complements team-based care and coordinates decisions across various systems and actors.	• Study human-AI teaming and multi-AI collaboration • Explore multi-modal AI for richer, more robust insights • Apply game theory to multi-stakeholder decision-making	• Seamless, coordinated care across providers • Fewer gaps or redundancies in care • Holistic decisions that consider the full patient context
AI Security and Privacy Protects the integrity of AI tools and patient data, thereby safeguarding trust.	• Defend against adversarial attacks • Invest in privacy-preserving AI (federated learning, differential privacy)	• Confidence that personal data is secure • Protection from malicious or faulty AI behavior • Willingness to engage with digital tools



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About ICF

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